Spatiotemporal Analysis of Davinci RNA-seq

- Pathway Biology of Normal Porcine Wound Healing:
  - Characterize Porcine genetic pathway activation underlying normal wound healing.
  - Identify biological programs associated with healthy porcine wound healing.
  - Leverage **Non-Negative Matrix Factorization** (NMF) to discover latent pathway activity.
  - **Fast Gene Set Enrichment Analysis** used to annotate NMF meta-genes.

Non-Negative Matrix Factorization Overview

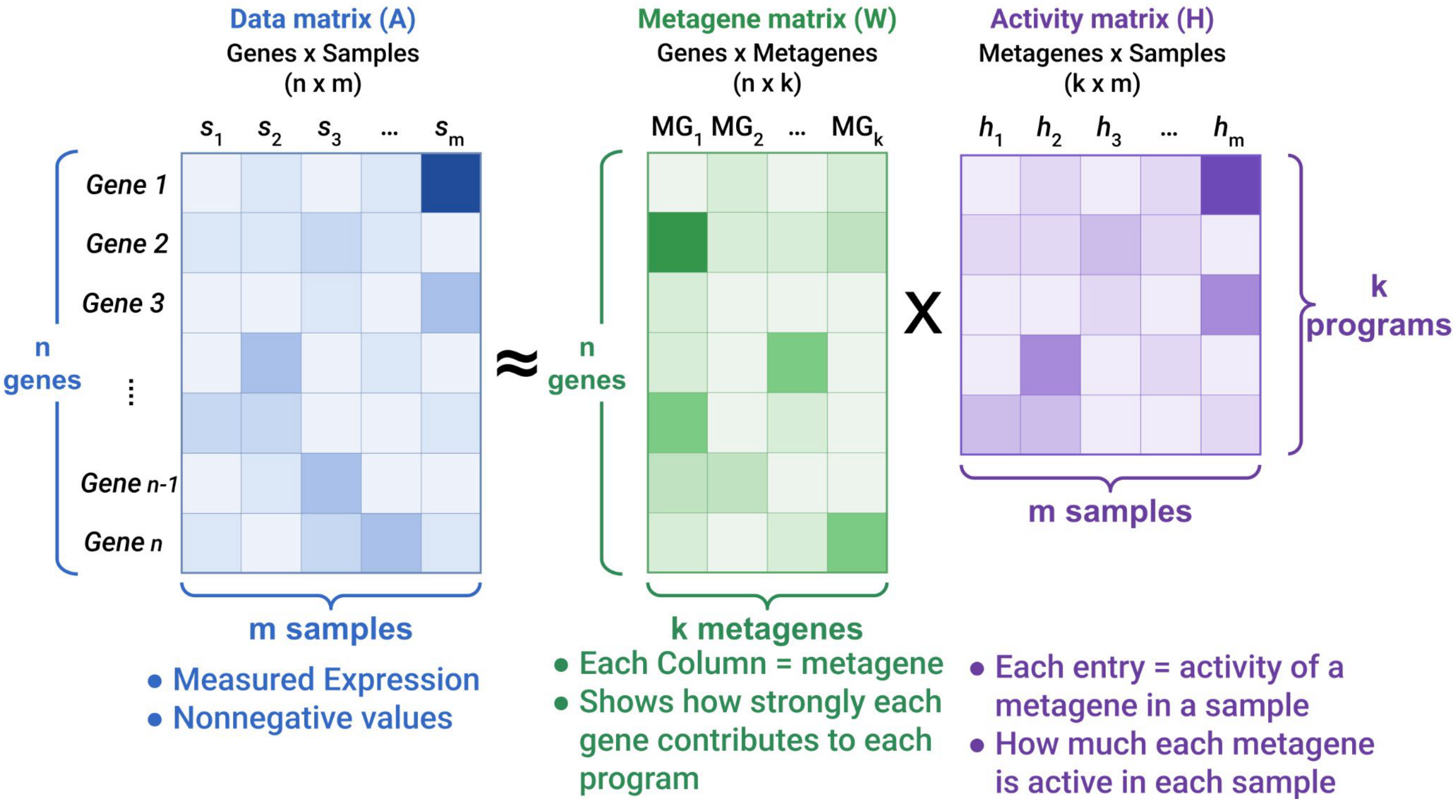

Figure 1: Overview of Non-Negative Matrix Factorization.

Non-Negative Matrix Factorization Results

- Healing-Specific Metagenes represent:
  - **Program 2:** Skin Morphogenesis & Development.
  - **Program 3:** Inflammatory Defense-Stress Response.
  - **Program 4:** Collagen, Cellular, Tissue Development.
  - **Program 6:** Cornification, Keratinization, Skin Barrier.

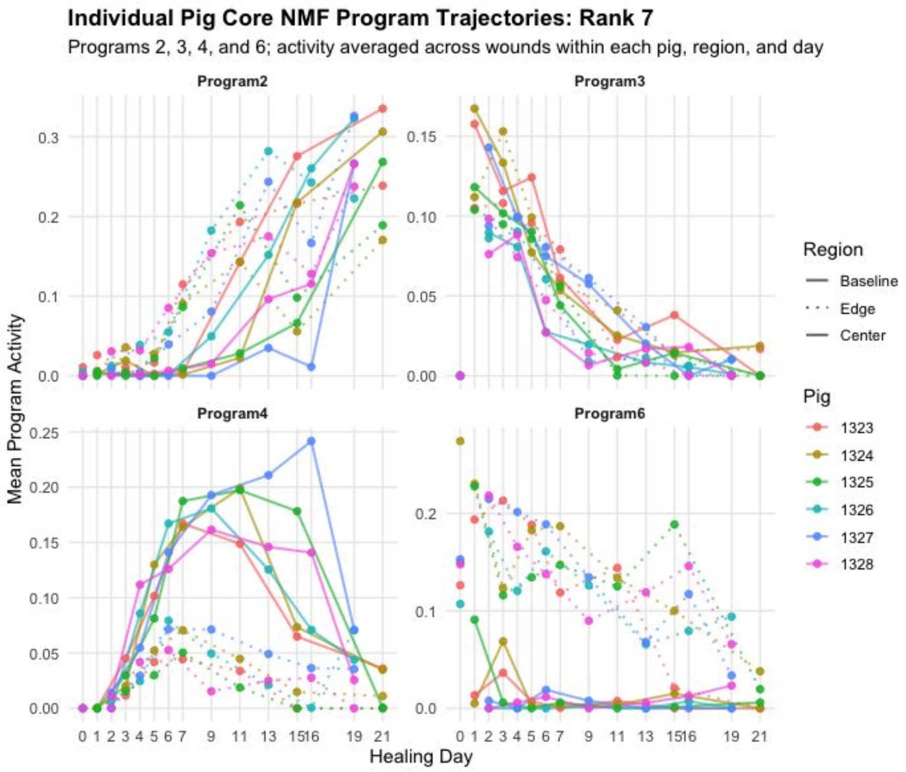

Figure 3: Temporal dynamics of 4 Healing-specific metagenes over 21 days.

PCA and UMAP of Healing-Specific Metagenes

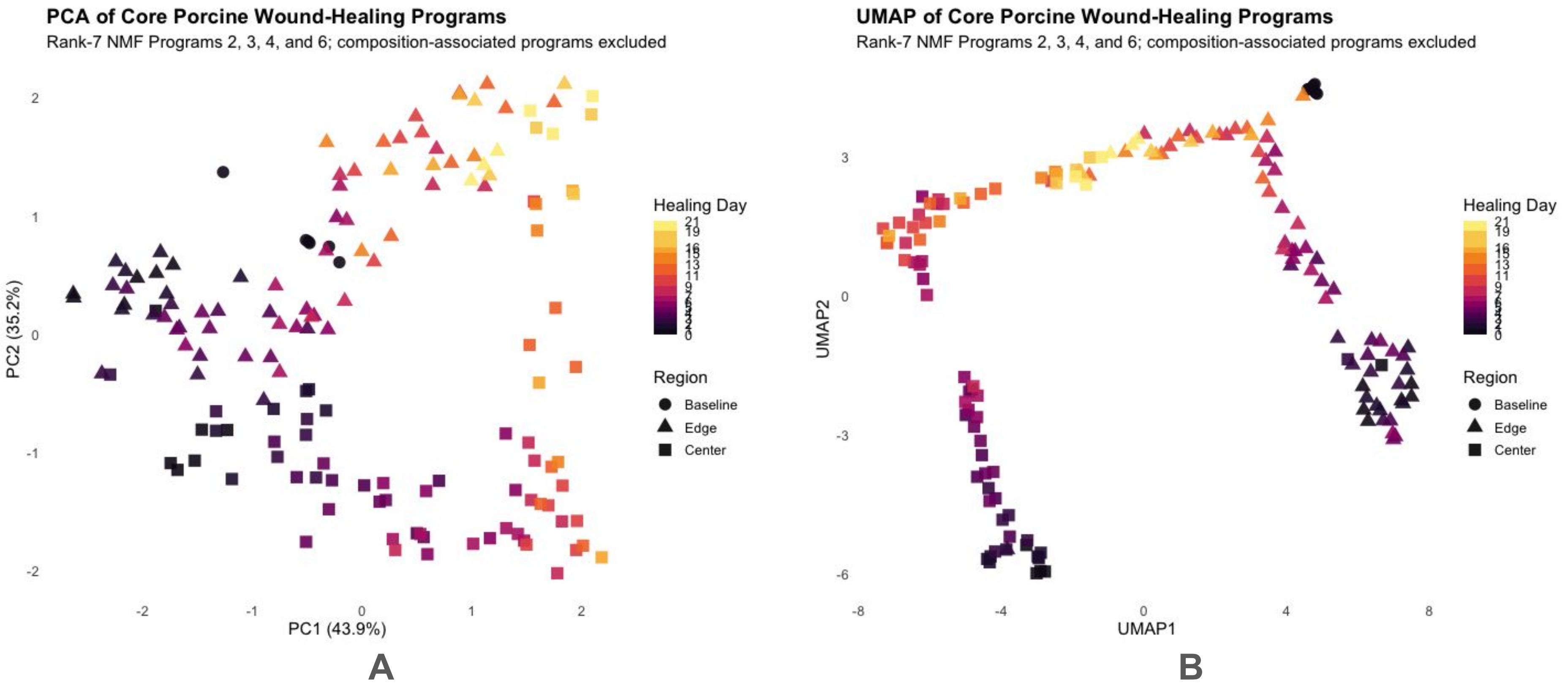

Figure 3: (A) PCA and (B) UMAP of four Healing-Specific metagenes over 21 days. NMF healing programs trace an organized trajectory through latent space, suggesting that wound repair can be represented by a compact set of molecular coordinates.